Venkata Lakshmi Parimala Pasupuleti

venkatalakshmiparimalap@gmail.com | +1-703-261-5410 | Fairfax, VA | LinkedIn | GitHub

SUMMARY

AI Engineer with 6+ years building production software systems and AI-powered applications. At Intuit, integrating LLMs (Claude, GPT, Llama) into enterprise financial workflows using the Anthropic Messages API, Agent SDK, and LangGraph -- RAG pipelines, prompt engineering, tool calling, vector-based retrieval, and agentic orchestration deployed on AWS. Built AgentProbe for cost-aware multi-provider LLM routing and LeaseI, an 8-agent RAG system (93.6% precision across 1,240 test cases). Experience with MCP (Model Context Protocol) and daily AI-assisted development (Claude Code). Strong Python, FastAPI, SQL, AWS, Docker, Kubernetes, and CI/CD. Background in REST API development, distributed systems, ETL pipelines, and responsible AI practices. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Python / FastAPI / SQL | LLM Integration (Claude, GPT, Llama) | RAG Pipelines / Vector Databases | Prompt Engineering / Tool Calling | Agentic Workflows (LangGraph) | Model Context Protocol (MCP) | ML Pipelines (scikit-learn, XGBoost) | AWS (Lambda, S3, RDS, SQS) | Terraform (IaC) | Docker / Kubernetes | CI/CD (GitHub Actions) | Observability (Prometheus, OpenTelemetry) | AI Evals / Golden Dataset Testing | REST APIs / Backend Systems | ETL Pipelines / Data Engineering | PostgreSQL / Advanced SQL

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- AI/LLM Tools & Production Systems -- Virginia, USA

- Build and deploy **LLM-based tools in a business context** using the Anthropic Messages API and Agent SDK -- prompt engineering, tool use, structured output, guardrails, and authn/authz for model plugins integrated into production financial data workflows.
- Own the **POC-to-production lifecycle**: prototype AI use cases, validate with golden-dataset eval harnesses, instrument with Prometheus and OpenTelemetry, and harden into enterprise-scale services on AWS.
- Build **ML pipelines** (scikit-learn, XGBoost) from feature engineering through scoring APIs; deploy via Docker and GitHub Actions CI/CD.
- Advise on technical guardrails, model cost management, and vendor evaluation for AI tooling; daily practitioner of AI-assisted development (Claude Code) with rigorous review before shipping.

Entrata -- Residential & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- Built and maintained **data-intensive backend systems** -- eligibility decisioning engine, ETL pipelines processing 1.6M+ records, async processing (RabbitMQ) with idempotency guarantees and dead-letter handling.
- Designed **Docker microservices architecture** (50% better scalability, 30% faster deployments); deployed on AWS with Terraform IaC; instrumented services for production observability.
- Owned **REST API integrations** with external federal data sources; optimized 200+ SQL queries via execution plan analysis (30% performance gain) across PostgreSQL/Oracle multi-schema environments.
- Built Python/Bash automation eliminating 80% of manual operational workload; reviewed 300+ PRs; mentored junior engineers on distributed systems and data pipeline patterns.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built backend modules (Python, Java, PostgreSQL) with data import pipelines processing 1.6M records; implemented role-based access control and multi-tenant data isolation.

PROJECTS

Leasel -- LLM-Based Compliance Tool (POC to Production)

Python · LangGraph · Agent SDK · FastAPI · RAG · PostgreSQL

8-agent LangGraph orchestration system demonstrating the full POC-to-production AI pattern: triage, RAG retrieval, rule evaluation, confidence scoring, and human escalation at defined thresholds. Observability and eval harness built in from day one. LLM-as-judge CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- AI Evaluation & Model Routing Infrastructure

Python · FastAPI · Provider Routing · Cost Management · CI/CD

15-endpoint FastAPI service with provider routing across five LLM/ML inference backends. Scoring engine evaluates output quality on five metrics with cost-aware routing; CI regression against golden datasets. Directly addresses AI platform observability, cost management, and flexible model adoption.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical NLP API

Python · Flask · REST APIs · RBAC · Docker

RESTful API translating clinical queries into FHIR data calls using LLM tools. Fine-grained RBAC, containerized deployment, and automated testing -- regulated-data AI integration pattern.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017